eVOLUCIÓN

Boletín de la Sociedad Española de Biología Evolutiva
Vol. 17(1) Marzo 2023



SESBE
Sociedad Española de Biología Evolutiva

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An interview with Erik Svensson

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Erik Svensson

Erik Svensson is Full Professor of Evolutionary Biology at Lund University, where he conducts empirical and theoretical work on various topics related to evolutionary biology. He is interested in the evolutionary dynamics of frequency-dependent processes, the maintenance of genetic polymorphisms, and the causal interplay between natural and sexual selection when it comes to producing reproductive isolation and, ultimately, speciation. In his empirical research, he mainly works with insects from the order Odonata (dragonflies and damselflies), though in the past he has worked with several other groups of organisms, including other insects, crustaceans, birds, and reptiles. Besides that, Erik also has a deep interest in the history of evolutionary thought, making him a rare case of a scientist with a wide knowledge not only of the basic theoretical toolkit of his discipline but also of its more profound conceptual foundations. I met Erik last year at Lund University, where I had the opportunity to talk with him about various scientific and philosophical topics. Erik kindly accepted my proposal of interviewing him for *eVOLUCIÓN*. Hope that you enjoy reading it as much as I did.

Q. Tell us a little bit about your background. When did your passion for the natural world arise? And how you eventually came to be interested in evolutionary biology?

R. Well, my background is similar to many other evolutionary ecologists of my generation, at least in Sweden, but differs

in some aspects. I was lucky to spend four years in southern Africa as a child, when my father worked for the Swedish International Aid Agency (SIDA), designing and planning schools in Botswana. I think it was there that I learned to appreciate natural history, particularly butterflies and frogs at an early age, but also larger mammals.

My interest in birds, spiders, and various other insect groups came later, when we moved back to Sweden. I joined the Swedish youth organization for naturalists ("Fältbiologerna", or the "Field Biologists") at the age of 13, and have been quite a passionate naturalist and photographer ever since. I collected insects as a teenager and was lucky to have a famous Swedish insect photographer (Åke Sandhall) as a mathematics (!) teacher in high school.

When I started studying biology, I discovered how mathematics was a useful tool and became fascinated with evolutionary biology, first by reading Stephen Jay Gould's books (particularly *Ever Since Darwin*) and somewhat later also Richard Dawkins' books (*The Selfish Gene, The Extended Phenotype* and *The Blind Watchmaker*).

Q. You were Barry Sinervo's first European postdoc student. Sinervo, who sadly passed away in 2021, had done outstanding work in several areas of evolutionary biology, from the impacts of environmental effects on fitness to the Rock-Paper-Scissors hypothesis for polymorphism maintenance. You collaborated on a recent piece in *Evolution* commemorating his work and legacy (Svensson et al. 2022). What things you learned from Barry, and what attributes did you find more inspiring in him?

R. I learned many things from Barry, and his laboratory was at its peak a very dynamic intellectual environment with many excellent and smart postdocs and PhD students, some of whom I still collaborate with to this day. Barry was not always easy to work with – I have to admit – and sometimes we disagreed profoundly, but he had an extremely creative mindset and was often able to "think out of the box", which forced me to do so as well. I was lucky to be in the same department (Department of Ecology and Evolutionary Biology at UCSC in California) at the same time as Barry and evolutionary geneticist and fruitfly biologist William Rice were both there. These two strong personalities really shaped my thinking and the way I do evolutionary biology. They represented two extreme ends of the continuum of our field: experimental evolution in the laboratory (Bill) and hardcore experimental evolutionary ecology in the field (Barry) and they often disagreed on scientific issues. But disagreement is not necessarily a bad thing – at least if you adopt a dialectical perspective of the world. Disagreements can be productive and force us

all to clarify and sharpen our arguments and become better scientists. Incidentally, it was Barry who suggested I should read Levins and Lewontin's *The Dialectical Biologist* (HUP, 1985) as well as Levin's classical work on evolution in changing environments, William Provine's books, etc. In short: Barry was a very integrative and broad-minded evolutionary biologist who thought deeply about the links between ecology, development, genetics, and the history of evolutionary biology.

Q. Throughout your career, you have worked with birds (blue tits), reptiles (side-blotched lizards), and insects (damselflies). What insights did you get from studying organisms with very different ecological roles and life histories? To what extent do the organisms chosen to conduct research determine the kinds of questions you can ask?

R. I think that organisms are extremely important and their importance is clearly underestimated when it comes to how research is interpreted by historians of science and biologists. The organism you choose to study determines what can be done (and what cannot be done!) and thereby also our direct focus of attention and interest. For instance, when I started working on lizards as a postdoc, a whole new world of thermal ecology opened up that I was not aware of before, having worked previously with endotherms like birds during my PhD. After only a few weeks of fieldwork with lizards at Barry's study site "Los Baños" in central California, I completely lost all my interest in parental care that I had been thinking about for five years as PhD-student working with blue tits!

I should also add that when I was an undergraduate student in Sweden, my ecologist teachers often told me that *"the organism should not matter, it is the questions"*. As a matter of fact, among Swedish ecologists during the 1970s, 1980s, and 1990s, there was often a negative attitude towards systematics, natural history, and evolution, based on this narrow mindset where "only the questions" was considered to be important. Thus, we were taught not to learn from the organisms and what questions were meaningful to ask from observing them in the field! Luckily, this has changed, especially as evolutionary biology became more firmly established in Sweden, after some initial resistance from conservative and traditionally-minded ecologists.

Q. In one of your recent papers (Svensson et al. 2020), you investigated how natural and sexual selection shape thermal phenotypic plasticity in certain odonate species in which there is a thermal conflict between fitness components, as survivorship is higher on low temperatures and reproductive

success increases with higher temperatures. Can you tell readers what were your main findings in this research?

R. We are very happy with this paper, as it provides a link between two important subfields that I am passionate about: thermal ecology and sexual selection. How do heat stress and increasing ambient temperatures affect insect body temperatures and ultimately their survival and reproductive success? How plastic are insect body temperatures and how is such thermal plasticity shaped by natural and sexual selection? To me, these are extremely important issues to address, both from a basic science viewpoint and in terms of the future of biodiversity on a heating planet. Our results show that temperature has different effects on survival and mating success, with higher temperatures increasing the number of days in which these damselflies can reproduce, but shortening their life-spans in the field. Also, we find that natural selection favors increased canalization of body temperatures, that is, individuals which are more thermally robust survive longer in the field. The results might indicate that insect life histories might change profoundly with increasing temperatures and high latitudes in Sweden, resulting in a shift towards a life history with high reproductive effort but shorter lifespans.

Q. Recently, a series of notable evolutionary biologists, including your Lund colleague Tobias Uller (see his interview for *eVOLUCIÓN*, 15(1), 2021: 54-63), have called for an Extended Evolutionary Synthesis (EES). You have been critical with these calls to extend or revise the Modern Synthesis (Svensson forthcoming). What are your thoughts on the EES? Does evolutionary theory really need a rethink?

R. I should clarify that my main issue with the EES is more how the history of evolutionary biology is described and the current characterization of our field, rather than any of the phenomena they highlight as being important, such as developmental bias, plasticity, niche construction, and non-genetic inheritance. In particular, the last of these phenomena is important and interesting and will clearly need more attention (indeed, it is increasingly getting such well-deserved attention from large parts of the evolutionary research community). But I think the EES advocates have somewhat underestimated both the scope and the current diversity of evolutionary biology as a field and they have also pushed a somewhat biased narrative of the field being more monolithic than it actually is, portraying the Modern Synthesis (MS) as having a more long-lasting influence than I think is warranted.

Nevertheless, I also have some issues with the so-called "traditionalists" who argue that "everything is fine" (e. g. Brian Charlesworth, Jerry Coyne) and who erroneously claim that not much has happened since the MS. So my take on this is that *both* the MS defenders and the EES proponents are somewhat mistaken; our field has changed a lot since the 1930s and 1940s, and will continue to do so, regardless of this debate which is largely about how to interpret the history. I have developed these critical points in more detail in a forthcoming book chapter (Svensson forthcoming) and in a popular science article in *The Conversation* (Svensson November 9, 2022).

Q. In a recent article for *Science for the People* commemorating Lewontin's dialectical materialism (Svensson November 3, 2021), you argue that, somehow, contemporary evolutionary biology and Marxism are natural bedfellows. Can you explain why do you think this is the case? What are your thoughts on the relationship between political ideology and science more broadly?

R. Actually, I did not argue that Marxism and evolutionary biology *are* natural bedfellows, but I raised the question *if* they are or could be so, based on the fact that so many leading evolutionary biologists and philosophers of biology were either explicit Marxists or had at least Marxist leanings. Examples of such biologists with Marxist leanings are J. B. S. Haldane and his student John Maynard Smith, Richard Levins, Richard Lewontin, Stephen Jay Gould, and philosopher of biology Elliott Sober. I think this is very interesting to note and it might indicate that dialectical thinking in Marxist philosophy can be a very productive tool also for evolutionary biologists. Actually, one does not even need to be a committed Marxist to see that. Other interesting examples of researchers who have clearly been influenced by Marxist philosophy are the theoretical evolutionary biologist Sergey Gavrilets and former population ecologist and nowadays quantitative historian Peter Turchin, both of whom were exposed to dialectical thinking in the Soviet Union, where they grew up and got their basic training.

Regarding the relationship between political ideology and science more broadly, I am tempted to say that ideology is false consciousness, which Karl Marx is supposed to never have said, but which some Marxists have attributed to him. That is, we must distinguish between the philosophical aspects of Marxism, and the particular political ideologies (socialism, communism) that emerged from Karl Marx ideas. In principle, also a liberal or bourgeois scientist holding a centrist or right-wing political view could appreciate dialectical thinking in Marxist philosophy,

without necessarily sharing the ideological beliefs or having sympathy for socialism or communism. More generally, however, I think scientists should realize that they are not objective or totally impartial observers of the world, but they have their ideological and political biases, whether they admit this or not. And in the age of a deep political crisis for humanity and threats of accelerating climate change and environmental destruction, maybe is the time for scientists to get out of their ivory towers and use their knowledge to actually try to change the world for the better.

Q. In 2012, you edited a volume devoted to the concept of Adaptive Landscape, commemorating the 80-year anniversary of Sewall Wright's classic paper on the topic (Svensson & Calsbeek 2012). In such volume you and Ryan Calsbeek state that "the Adaptive Landscape idea must be one of the most successful concepts in evolutionary biology". What are your arguments to hold such a view? What challenges lie ahead for the concept of Adaptive Landscape, given recent advances in evolutionary theory?

R. I need to slightly correct the quotation from our last chapter in this edited volume. We stated: *"**If** a scientific metaphor or concept such as the Adaptive Landscape should be judged by the number of studies and publications that has resulted from it, the Adaptive Landscape must surely be one of the most successful ideas in evolutionary biology…"*. Note the crucial "if"! I am not sure whether the number of publications or studies that have been stimulated by a concept or a theory is – or should be – a criterion for whether the concept or theory was useful (or not). But clearly, the Adaptive Landscape has stimulated a lot of research, and it is up to future historians of science and biologists to judge whether all this work was well-spent time or whether it was just chasing a misleading idea.

Personally, I think the Adaptive Landscape concept has been extremely important in evolutionary theory, and there are whole subfields that probably would not have existed without it. Examples include experimental evolution studies focusing on contingency vs. determinism and much of the work on ecological speciation. I also think that some philosophers and biologists conflate the visual metaphor of the Adaptive Landscape – from Sewall Wright and onwards – with the actual formal mathematical and statistical framework that has been developed by Russell Lande and colleagues from the 1970s and onwards.

Regarding the future of the Adaptive Landscape, I think it will remain with us for a long time, and there is currently a lot of interesting work going on in various

new areas, such as metabolic networks (e.g. Andreas Wagner's work) and experimental evolution studies on microbes addressing the importance of gene interactions. So, I think the Adaptive Landscape will survive at least another 80 years, as we wrote in our 2012 volume that celebrated the 80 years that had passed since Sewall Wright's original publication in 1932 when this concept was first formulated. Thus, I fundamentally disagree with the philosophers Jonathan Kaplan and Massimo Pigliucci who prematurely declared the Adaptive Landscape to be dead in a couple of papers in *Biology & Philosophy* from the 2000s. They underestimated the power of this concept, were too much in a rush, and were both deeply mistaken, in my opinion. It is probably no coincidence that empiricists like myself are more positive about the Adaptive Landscape as an organizing research framework than philosophers who are not actively involved in empirical evolutionary research programmes.

Q. Tell us a little bit about your research lines right now. What are you working on? What are the main insights you expect to gain on how evolution works from your current project/s?

R. I am currently fascinated about evolutionary predictability and its limits: when it works, when it breaks down, and on what time scales. We have a relatively successful theoretical machinery built on evolutionary quantitative genetics that grew out of plant and animal breeding, and that has sometimes (although not always) been successful in when applied to natural populations. There is also interesting ongoing work on how to integrate quantitative genetics with phylogenetic comparative methods, and thereby on how to build bridges between micro- and macroevolution. So far, so good, but there are difficulties with this promising project that are interesting to understand and explore.

In particular, higher-level population and species processes such as biased extinction and speciation will partly decouple macroevolutionary patterns from microevolutionary processes. It is important to understand such decoupling, how and when it happens, and to what extent it affects evolutionary predictability. I explore this topic using long-term population data from 22 years of natural field populations of the common bluetail damselfly (*Ischnura elegans*) with my students and postdocs and also by combining this work with interspecific data and phylogenetic comparative analyses to explore and better understand both micro- and macroevolutionary dynamics of Odonata (dragonflies and damselflies). We are particularly interested in how temperature can affect evolutionary predictability at various time scales.

Q. You have a wonderful collection of nature photos (available on your Instagram account, @eriksvenssonsweden), and you describe yourself as a proud naturalist who is not ashamed of loving nature, birds, insects, and beauty. However, the traditional image of scientists is one of detached observers of reality, who should not be biased or influenced by their feelings. What are your thoughts on such an image? How do you reconcile your views as a scientist and as a naturalist in your work and daily life?

R. I think there is actually somewhat of a tension between being a naturalist and being a quantitative and rigorous scientist. But such tension is not necessarily bad, at least not always so, especially if one is aware of it. My entrance into biology was not initially because of my interest in theoretical questions in ecology and evolution, but more from a natural history background, being fascinated by biodiversity and the beauty of our natural world since my childhood years. That fascination and sense of beauty is nothing I am ashamed of. It has also been useful – at least during some periods – in the sense that it has helped me to find interesting study systems in nature that maybe not everyone would find. However, it also means that by some I might be perceived as "only" a field biologist and evolutionary ecologist. But I am not only a naturalist, as I am also interested in many conceptual issues in evolutionary biology and I read – with pleasure and interest – much research that I would never like to do myself, e. g. experimental evolution studies on *Escherichia coli* or *Drosophila melanogaster*. I learn a lot from such interesting work, but I would probably not enjoy doing it myself (although in collaboration with others I am fine working on such systems). Instead, I see myself as a "bridge builder" between evolutionary ecologists working in the field, population genetics, theoretical evolutionary biologists, and experimental evolution communities. And that is not a bad position to be in, I must say.

Q. What advice would you give to a young scholar interested in evolutionary biology? What are the things you consider more important to understand how evolution works?

R. I would like to reiterate what my esteemed Lund colleague Tobias Uller said when he was interviewed in this same series as you are now interviewing me: *"Do what you find interesting"*. I think this advice is most of what you need. You should not do what others expect you to do, or because of external pressure on what is fashionable or what you perceive as "strategic", but your motivation must come from yourself. If you do what you are interested in, hopefully, enough of your colleagues will find it interesting too, and in the long run, you will be happier and intellectually satisfied.

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